

# 29.9% of returns aren't genuine. They carry **\$4.64M** of refund dollars — and now we know which ones.

A retailer's return-audit team reviews every disputed return the same way — spreadsheet speed, inconsistent criteria. We rebuilt **60,000 orders** into a PySpark/MLlib pipeline that triages each one into legitimate, policy abuse, fraud, or wardrobing, priced what's actually at stake, and separately segmented customers to surface the product-quality problem no fraud label can see.

## NON-LEGITIMATE REFUND EXPOSURE

# \$4.64M

**43.8%** of all refund dollars requested, from just **29.9%** of orders — bad actors skew toward higher-value claims

## CLASSIFIER PERFORMANCE (TEST SET)

# 0.999

weighted F1, Random Forest — synthetic-dataset caveat on slide 7, not a production guarantee

## HIDDEN IN THE "LEGITIMATE" MAJORITY

# 38%

of returns in the largest, 96%-legitimate customer segment are still quality-issue driven

**60,000 orders · 35 features**, Kaggle e-commerce return-abuse dataset · Fraudulent Return claims average **\$434.73** vs. **\$141.14** for legitimate returns — a **3.1x** gap · three independent PySpark MLlib approaches: multi-class classification (Random Forest vs. Logistic Regression), K-Means clustering, and refund-exposure regression · this dataset is synthetic, so treat 0.999 F1 as a pipeline proof, not a production number (more on slide 7).

## THE BUSINESS PROBLEM

# One review queue, *three different problems* wearing the same form.

A return-audit team currently reviews every disputed return the same way. It has no systematic way to tell whether a spike in returns is fraud, abuse of a lenient policy, or a genuine quality defect — three problems that need three different owners, currently indistinguishable in a spreadsheet.

### 01 · CUSTOMERS RETURN

#### Most returns are genuine

Defective items, wrong fit, changed mind — the normal cost of running an e-commerce store.

**70.1%** legitimate

### 02 · SOME ABUSE IT

#### Policy abuse, fraud, wardrobing

Exploiting lenient terms, deceptive claims, or wearing-and-returning — three distinct behaviors.

**29.9%** of orders

### 03 · ALL LOOK THE SAME

#### One manual-review process

Every disputed return gets identical treatment today, regardless of which of the three it actually is.

**1** review queue

### 04 · THE WRONG FIX

#### Blanket policy tightening

Punishes the 70% of legitimate customers to catch the fraction who abuse the system.

**wrong lever**

## THE QUESTION THIS PROJECT ANSWERS

*Which returns are fraud, which are policy abuse, and which are genuine quality failures — and what does each one actually cost?*

Three sub-problems, one dataset: **return fraud** → classification on abuse\_type; **product quality** → return-reason patterns cut by cluster and category; **return risk** → lifetime order/return history per customer.

## WHY THIS IS A TRIAGE PROBLEM, NOT A POLICY PROBLEM

The highest-value lever is routing each return to the right team fast — not tightening policy across the board, which would punish the 70% of customers who did nothing wrong. **Three different interventions, three different owners**, is the recommendation this whole pipeline builds toward.

## INDUSTRY BACKGROUND

# This is not a hypothetical problem — *the industry is already living it.*

E-commerce carries a structurally higher return burden than brick-and-mortar, and fraud/abuse is a large, growing share of it. This project sits squarely inside an industry-wide shift already underway.

### NRF · 2025 RETAIL RETURNS LANDSCAPE

# 19.3%

of online sales are returned — above the 15.8% all-channel average. \$849.9B in merchandise returns forecast for 2025.

### APPRISS RETAIL × DELOITTE · 2024

# 15.14%

of all returns are fraudulent — \$103B lost in 2024 (up from \$101B in 2023), across 60+ retailers studied.

### SAME 2024 STUDY

# 60%

of retail executives specifically flag **wardrobing** — one of our four target classes — as a major fraud pattern. 85% now deploy AI to fight it.

### HOW OUR DATASET COMPARES

This dataset's combined Fraudulent Return + Wardrobing rate is **10.2% + 7.7% = 17.9%** — close to Appriss's industry-wide **15.14%** fraud rate. The class **prevalence** here is realistically calibrated to real-world reporting, even though (see slide 7) the class **separability** the model achieves is not something to expect on live data.

Sources: National Retail Federation, *2025 Retail Returns Landscape* (nrf.com) · Appriss Retail with Deloitte, *Fraudulent Returns and Claims Cost Retailers \$103B in 2024* (apprissretail.com). Full citations in the report's Appendix C.

DATA OVERVIEW · 60,000 ORDERS, 35 FEATURES

# A realistic imbalance — and the dollars don't split the way the order counts do.



■ Legitimate — 70.1% (42,060 orders, \$5.94M requested) ■ Policy Abuser — 12.0% (7,192, \$0.76M) ■ Fraudulent Return — 10.2% (6,112, \$2.66M) ■ Wardrobing — 7.7% (4,636, \$1.22M)

## Behavioral signals

Photo evidence, tracking validity, address changes, refund-to-different-account, multiple accounts, dispute history

## Customer history

Lifetime orders/returns, return rate, account age, loyalty segment

## Context

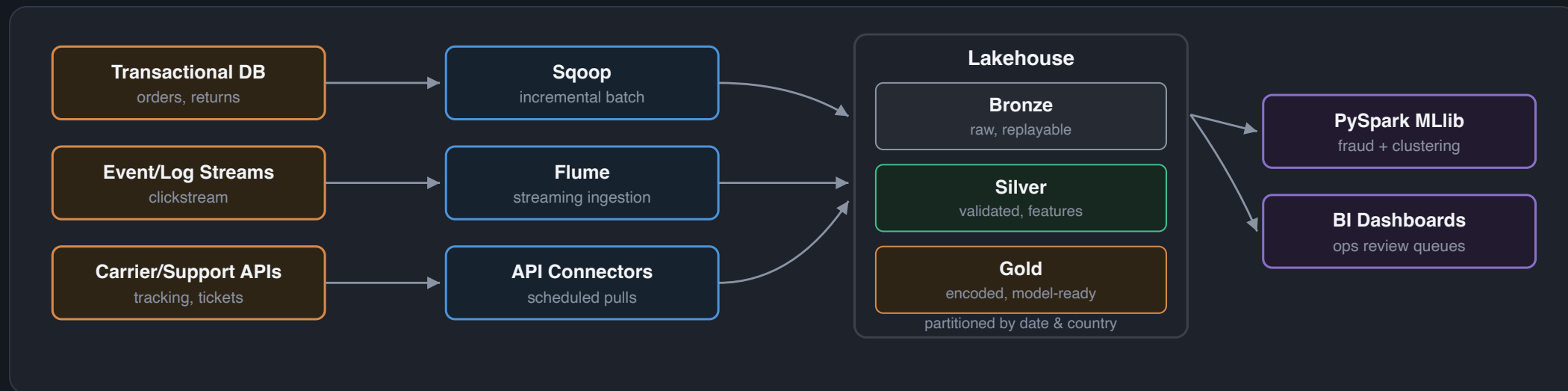
12 product categories, 12 return reasons, 8 countries, payment method, shipping carrier

**VERIFIED — CHECKED PROGRAMMATICALLY, NOT ASSUMED**

**0 missing values** across all 35 columns, **0 duplicate** order\_id, **0 logical violations** (no return-before-order dates, no returns exceeding lifetime orders, no implausible ages) — confirmed in notebooks/01\_data\_engineering.ipynb §2 before anything downstream trusted this data.

# Sqoop / Flume / API ingestion → *Bronze/Silver/Gold lakehouse* → PySpark MLlib.

This notebook runs directly on the static Kaggle CSV as a practical stand-in — the diagram below is the production-scale design the Silver→Gold transformation logic is built to generalize to.



## WHY BRONZE/SILVER/GOLD

Bronze keeps a permanent, replayable raw record. Silver is cleaned and feature-engineered once, reused by both models. Gold is the only layer anything ML touches.

## EXPLICIT SCHEMA, NOT INFERRED

The notebook loads the CSV with an explicit StructType — never inferSchema in production, which forces a full extra scan and can silently mis-type columns on drift.

## CAP OUTLIERS, DON'T DELETE

Extreme values are often the fraud signal itself. 8 continuous fields IQR-capped at the 1st/99th percentile — preserving row count and the extremeness of the behavior.

# What the three models actually tell us — *and the risks that travel with them.*

## KEY FINDINGS

- **Non-legit returns skew expensive, not just frequent.** 29.9% of orders carry 43.8% of refund dollars (\$4.64M) — fraud claims average 3.1× a legitimate return.
- **The classifier earns its complexity.** Random Forest hits perfect precision/recall where a free 4-flag heuristic tops out at F1 0.61.
- **The quality problem is broad-based, not one vendor.** Quality-issue rates band 31.5%–35.2% across all 12 categories — a systemic pattern, not a vendor hunt.
- **A simpler model wins honestly on refund pricing.** Linear Regression beats both a naive baseline (40% RMSE) and a more complex GBTR regressor.
- **Our dataset's fraud rate is industry-realistic.** 17.9% (Fraud + Wardrobing) sits close to Appriss Retail's reported 15.14% — the scale here is real.

## RISKS & FUTURE ROADMAP

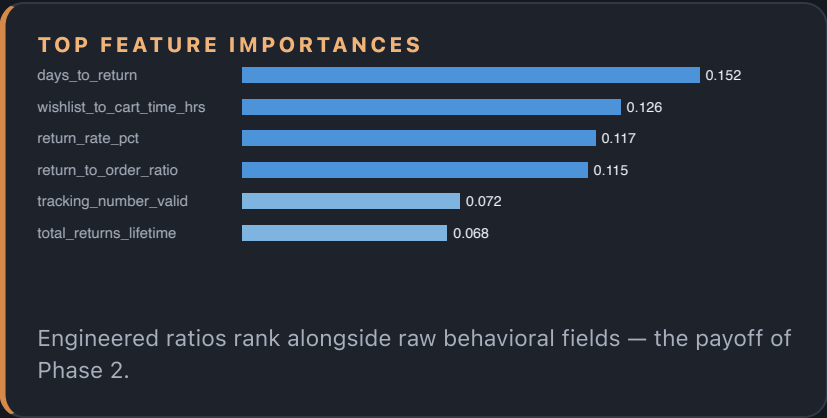
- **Risk — synthetic-data overfit.** Treat this model as a pipeline proof-of-concept; re-train on real labeled outcomes before go-live, budgeting for 80–90% F1, not 99.9%.
- **Risk — no production ground-truth labels.** Real retailers rarely have a clean `abuse_type` column — build human-in-the-loop labeling from the review queue this system creates.
- **Risk — the quality finding explains *what*, not *why*.** Broad-based isn't root-caused; that needs a packaging/supplier QA audit this dataset can't run.
- **Roadmap.** Real-time scoring at return-request time, a feedback loop from reviewer overrides back into re-training, and a rolling time-based validation window once live data exists.

# Random Forest vs. Logistic Regression on *return-fraud type*.

Spark MLlib's `GBTClassifier` only supports binary classification — not usable for this 4-class target — so Random Forest, which natively supports multi-class, is the primary model. Class-weighted training (inverse frequency) handles the 70/12/10/8 imbalance; evaluation reports per-class precision/recall/F1, not headline accuracy.

| MODEL                          | WEIGHTED PRECISION | WEIGHTED RECALL | WEIGHTED F1 |
|--------------------------------|--------------------|-----------------|-------------|
| Logistic Regression (baseline) | 0.998              | 0.998           | 0.998       |
| Random Forest (final)          | 0.999              | 0.999           | 0.999       |

200 trees, maxDepth=10, class-weighted. Train 45,112 / test 14,888 rows, 75/25 split, seed=42.



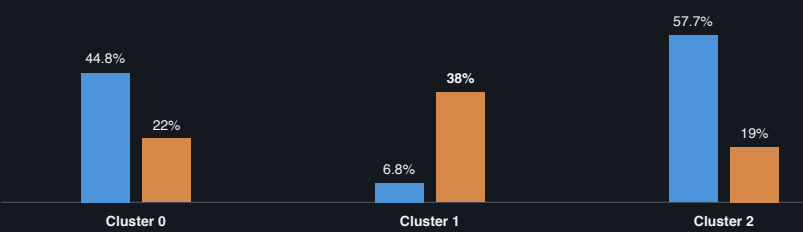
**Honest note:** this is a Kaggle synthetic dataset, so 0.999 F1 is higher than we'd expect in production — feature importance is spread across ~15 features with no single leak, but a realistic target on live data is **80–90% F1**. We treat this model as proof of the pipeline, not a performance guarantee. Full discussion in `docs/fraud-classification-results.md`.

APPROACH 2 — CLUSTERING

# K-Means finds 3 behavioral segments *the fraud label can't see.*

An independent, unsupervised lens: features deliberately exclude the fraud-flag columns used by the classifier, so clustering surfaces behavior/quality patterns a supervised label alone can't. k=3 selected by silhouette score over k=2..6.

| CLUSTER                   | N      | AVG RETURN RATE | AVG ORDER VALUE | % QUALITY-ISSUE | DOMINANT MIX            |
|---------------------------|--------|-----------------|-----------------|-----------------|-------------------------|
| 0 — High-value fraud-risk | 6,646  | 44.8%           | \$484           | 22%             | Fraudulent Return-heavy |
| 1 — Core legitimate base  | 43,829 | 6.8%            | \$157           | 38%             | 96% Legitimate          |
| 2 — Frequent policy-abuse | 9,525  | 57.7%           | \$129           | 19%             | Policy Abuser-heavy     |



KEY INSIGHT

Cluster 1 has by far the **lowest** return rate but the **highest** quality-issue share — the two metrics move in opposite directions, which is exactly why a single blended "return rate" KPI hides this segment's problem from whoever owns it today.

This is a **QA/vendor problem**, not a fraud problem — cross-reference dominant product\_category within Cluster 1 against return\_reason to target the escalation.

# Four interventions, four owners — *ship the free one first.*

## SHIP THIS FIRST, NO ML REQUIRED

Require photo evidence above a value threshold. The "high-value item + no photo evidence" pattern alone touches **\$1.88M** of non-legitimate refund exposure across **4,673 orders** — a policy rule, live immediately, independent of the classifier's rollout timeline.

### 01 Deploy the fraud classifier as a triage score, not an auto-block.

Route the highest `fraud_signal_score` decile + Cluster 0 to manual review; auto-approve the lowest-risk tail. **Re-threshold asymmetrically** — a false "Legitimate → Fraudulent Return" flag costs more in trust than a slow review costs in ops time.

OWNER · FRAUD & RISK

### 02 Give Cluster 1's 38% quality-issue rate to QA as a systemic audit, not a vendor hunt.

Quality-issue rates are broad-based — banded 31.5%–35.2% across all 12 categories, not one bad vendor. **73% of customers**, near-zero fraud, but the highest quality complaint rate of any segment.

OWNER · QA / VENDOR MANAGEMENT

### 03 Tighten policy — don't investigate — for Cluster 2.

High order/return volume, low order value, Policy-Abuser-dominated. A shorter return window or restocking fee targets this pattern more precisely than a fraud investigation would.

OWNER · CUSTOMER POLICY

### 04 Feed the refund-exposure regressor to Finance, not just Fraud & Risk.

Linear Regression prices expected refund dollars the moment a return is filed — a reserve-accounting input independent of whether it turns out to be fraud, abuse, or genuine.

OWNER · FINANCE / CUSTOMER OPS



THANK YOU

# Thank you. Questions *welcome.*

We'd rather be asked about the caveats than the number —  
**\$4.64M** of non-legitimate refund exposure, **\$1.88M** of it  
reachable today with a single no-ML policy rule.

Subham Ranjan · A017

Shriya · A023

Abhinav Kumar · A035

Dev Vagrecha · A049

Samruddhi Pradhan · A054

Yatharth Vij · A072

## IN THE REPOSITORY

notebooks/ — two executable  
PySpark notebooks, Bronze to  
models

## WRITTEN UP

docs/ — per-model results, each  
with its own weaknesses section

## THE DELIVERABLES

deliverables/ — this deck and the  
full consulting report, HTML + PDF

## REPRODUCIBILITY

seed=42 throughout · every number  
here read from the notebooks' own  
output

Dataset: `ecommerce_return_abuse_dataset.csv` (Kaggle) · references, AI-usage disclosure and team contributions in the consulting report's Appendix.